Self-Similar Recursive Grids: Formalization, Symmetry, and Learning Rules Beyond Backpropagation

Working notes on the "recursive grid" idea

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Abstract

We formalize the "recursive grid" idea. A grid of n cells has a configuration C and an outside state $f(C)$. Grids are the cells of larger grids, and one cell map f is used at every position and every scale. We call the result a *self-similar recursive grid* (SRG). Structurally, an SRG is a recurrent network unrolled in *scale* instead of time, with branching n. This one observation organizes everything else. An RNN has a single trajectory; an SRG has n^L root-to-leaf trajectories and $N_L = (n^L - 1)/(n - 1)$ copies of the same local problem per input.

We prove a set of results that say exactly which non-backpropagation learning rules the symmetry makes possible, and when they win: (i) the invariance group of an exchangeable SRG is exactly the iterated wreath product $\mathrm{Aut}(T_L)$, and an SRG with a multiset-injective cell map is a complete invariant for it; (ii) every $\mathrm{Aut}(T_L)$-invariant covariance has exactly $L{+}1$ eigenvalues, one per scale, so linear learning dynamics decouple by scale and the optimal preconditioner is $L{+}1$ numbers; (iii) a discrete SRG is a tree automaton, and a Myhill–Nerode theorem gives its minimal state space; the "copy grids with equal histories" idea is exactly this minimal quotient, and scale tying is precisely what makes cross-scale ("diagonal") copying valid; (iv) one constant κ (the scale gain of f) bounds both how copy errors are amplified and how much each level contributes to the backprop gradient; in the stable regime $\kappa < 1$, backprop trains the shared map almost entirely from the few top-level nodes; (v) for self-similar generative processes, Bayes-optimal inference *is* an SRG, the shared kernel is identifiable from the law of a single bottom grid, the latent labelling is pinned by self-similarity itself (so no labels are needed for generic kernels), and one unlabelled example supplies n^{L-1} conditionally independent samples of the kernel; (vi) three concrete local rules come with guarantees: frozen-input ("scale-equilibrium") learning, scale-Bellman bootstrapping with error linear in depth, and node-parallel evolution strategies, which have a population of size N_L at the cost of two forward passes. We check every quantitative claim numerically at toy scale. In the headline demonstration, a single unlabelled tree with 19,683 leaves yields a root classifier within 0.2 points of the Bayes oracle, with zero labels and no backpropagation.

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1 The object, cleaned up

1.1 Trees and heights

Fix a branching factor $n \geq 2$ and a depth $L \geq 1$. The complete rooted n-ary tree T_L has node set $V = \bigcup_{\ell=0}^{L} [n]^\ell$ (words over $[n] = \{1, \ldots, n\}$), root ϵ, and children $w1, \ldots, wn$ of each node w with $|w| < L$. The *height* is $\mathrm{ht}(w) = L - |w|$, so leaves have height 0 and the root has height L. Write $V^{(h)}$ for the n^{L-h} nodes of height h, and $N_L = \sum_{h=1}^{L} n^{L-h} = (n^L - 1)/(n-1)$ for the number of internal nodes (grids). The leaves below a node w form its *block*.

1.2 Self-similar recursive grids

Definition 1.1 (SRG)**.** A *self-similar recursive grid* is a tuple $(\Sigma, \mathcal{X}, e, f, r)$ with an input alphabet Σ, a state space $\mathcal{X}$, a leaf encoder $e : \Sigma \to \mathcal{X}$, a *cell map* $f : \mathcal{X}^n \to \mathcal{X}$, and a readout (observer) $r : \mathcal{X} \to \mathcal{Y}$. Given an input $\sigma : [n]^L \to \Sigma$, define

$$x_w = e(\sigma_w) \ \ (|w| = L), \qquad C_w = (x_{w1}, \ldots, x_{wn}), \qquad x_w = f(C_w) \ \ (|w| < L).$$

C_w is the *configuration* of grid w, x_w its *outside state*, and $r(x_w)$ what the observer reports about grid w. The same f is used at every position and at every height.

Dictionary to the original notes:

notes	formal object
percepts of grid i	cells $x_{w1}, \dots, x_{wn}$
configuration C_i	$C_w \in \mathcal{X}^n$
outside state $f(C_i)$	$x_w = f(C_w) \in \mathcal{X}$
"f maps C_i to a coordinate in the larger space"	$\mathcal{X}$ (lattice, $\mathbb{R}^d$, simplex, finite set)
process P_i	the law (and dynamics) of the leaf input σ
observer $F(S, C_i)$, strategy S from learner $\mathcal{L}$	readout r (and f), chosen by a learning rule
history $H(G_i)$	$H_w(t) = (C_w(t'))_{t' \le t}$ (§3.1)
"the substrate is identical"	one f at all N_L nodes (scale tying)

Height maps. Let $F_0 = e$ and $F_{h+1}(s) = f\big(F_h(s_1), \dots, F_h(s_n)\big)$ for $s \in \Sigma^{[n]^{h+1}}$, where s_i is the i-th sub-block of s.

Lemma 1.2 (self-similarity)**.** *$x_w = F_{\mathrm{ht}(w)}(\sigma|_w)$, where $\sigma|_w(u) = \sigma_{wu}$ is the content of w's block. In particular, every node of height h, at every position, computes the same function F_h, and $F_{h'}$ contains $n^{h'-h}$ copies of F_h as subcomputations whenever $h \le h'$.*

Proof. Induction on height. □

Variants.

1. *Scale-untied*: height-dependent maps f_h (weights still shared across positions). We use this only as a comparison point.
2. *Exchangeable*: $f(c_{\pi 1}, \dots, c_{\pi n}) = f(c)$ for all $\pi \in S_n$. The percepts of a grid then form an unordered set.
3. *Lattice (the "simpler cousin")*: $n = b^d$ and child i is an offset in $\{0, \dots, b-1\}^d$. With $f(c) = \phi(\sum_i W_i c_i + \beta)$, an SRG is *exactly* a depth-L CNN with kernel size b and stride b that uses the same weights at every layer. Grid w of height h is a $b^h \times \cdots \times b^h$ patch, and its "coordinate in the larger grid" is the point $\lfloor p/b^h \rfloor$ of the coarse lattice, where p is any leaf position inside the patch.
4. *Temporal*: $x_w(t) = f(C_w(t), x_w(t-1))$ with leaf inputs $\sigma(t)$ (§3.1).

1.3 The structural fact: an RNN in scale

Proposition 1.3. *The SRG is the depth-L unrolling of a one-vertex graph with n directed self-loops. For $n = 1$ it is the iterated map $x_L = f^{\circ L}(e(\sigma))$, i.e. an autonomous RNN unrolled for L steps. For $n \ge 2$, the unrolling has n^L root-to-leaf paths ("scale trajectories"), and the same map is applied at N_L nodes.*

The whole theory of recurrent networks therefore has scale analogues, with *time replaced by scale*. There is one crucial difference: an RNN sees *one* trajectory, while an SRG sees an exponentially large *ensemble* of trajectories that share the map.

RNN (time)	SRG (scale)	where
backprop through time	backprop through scale	§4
exploding/vanishing gradients	scale gain κ; criticality	Prop. 4.2
TD / Bellman bootstrapping	scale-Bellman bootstrapping	§6.2
causal states / minimal automata	Myhill–Nerode for grids	§3
one trajectory	n^L trajectories, N_L copies of f	§5,§6

2 Symmetry

Two different symmetries are easily conflated, so we separate them. *Computational self-similarity* (Lemma 1.2) holds for every SRG: the parameter f has a single orbit, namely all nodes. *Functional invariance* holds only for exchangeable SRGs: a group acts on inputs and leaves F_L unchanged. The learning rules in §6 mostly use the first; the spectral results below use the second.

2.1 The invariance group

Let $\mathrm{Aut}(T_L)$ act on leaves (permutations induced by tree automorphisms). Choosing a permutation of the children at each internal node gives

$$\mathrm{Aut}(T_L) \cong \underbrace{S_n \wr S_n \wr \cdots \wr S_n}_{L}, \qquad |\mathrm{Aut}(T_L)| = (n!)^{N_L}.$$

A leaf permutation lies in $\mathrm{Aut}(T_L)$ iff it maps every block to a block.

Proposition 2.1. *If f is exchangeable, then $F_L(\sigma \circ g) = F_L(\sigma)$ for every $g \in \mathrm{Aut}(T_L)$.*

Proof. Induction: an automorphism permutes children and applies automorphisms inside each child's subtree. The children's states are invariant by induction, and f ignores their order. □

Proposition 2.2 (exactness and completeness)**.** *Let $\Sigma \supseteq \{0,1\}$, let e be injective, and let f be* multiset-injective *on reachable states: $f(c) = f(c')$ iff c' is a permutation of c. Then*

1. *$F_L(s) = F_L(s')$ iff $s' = s \circ g$ for some $g \in \mathrm{Aut}(T_L)$ (complete invariant);*
2. *$\{g \in \mathrm{Sym}(\textit{leaves}) : F_L \circ g = F_L\} = \mathrm{Aut}(T_L)$ exactly;*
3. *every $\mathrm{Aut}(T_L)$-invariant function factors as $\tilde{\phi} \circ F_L$ (universality).*

Proof. (1) Induction on height. If $F_{h+1}(s) = F_{h+1}(s')$, multiset-injectivity gives a permutation π with $F_h(s_i) = F_h(s'_{\pi i})$. By induction $s_i = s'_{\pi i} \circ g_i$, and π together with the g_i is an automorphism. (2) Suppose $g \notin \mathrm{Aut}(T_L)$. Then some block B (of height $1 \le h < L$) has an image $g(B)$ that is not a block. Put $\sigma = \mathbf{1}_B$. If $F_L(\sigma \circ g^{-1}) = F_L(\sigma)$, then by (1) $\mathbf{1}_{g(B)} = \mathbf{1}_{a(B)}$ for some automorphism a, so $g(B) = a(B)$ is a block, a contradiction. (3) follows from (1). □

Such f exist on the countable reachable set of any finite Σ. Continuous injective f cannot exist on all of $(\mathbb{R}^d)^n$, for dimension reasons. *Checked numerically for $(n, L) \in \{(2,2),(2,3),(3,2)\}$ by enumerating all $(n^L)!$ leaf permutations: the stabiliser orders are $8, 128, 1296 = (n!)^{N_L}$.*

2.2 Scale decomposition: the symmetry leaves exactly $L+1$ numbers

For $0 \le k < L$, let U_k be the functions on leaves that are constant on height-k blocks and sum to zero on every height-$(k{+}1)$ block. Let U_L be the constants. These are the tree-Haar (wavelet) detail spaces: $\mathbb{R}^{n^L} = \bigoplus_{k=0}^{L} U_k$ orthogonally, with $\dim U_k = (n-1)n^{L-k-1}$ and $\dim U_L = 1$. Let $h(i,j)$ be the height of the lowest common ancestor of leaves i and j.

Theorem 2.3 (scale spectrum)**.** *Let C be an $n^L \times n^L$ matrix with $C_{gi,gj} = C_{ij}$ for all $g \in \mathrm{Aut}(T_L)$. Then $C_{ij} = c_{h(i,j)}$ for some numbers $c_0, \dots, c_L$, and C acts on U_k as the scalar*

$$\sigma_k = \sum_{h=0}^{k} N'_h c_h - n^k c_{k+1} \quad (0 \le k < L), \qquad \sigma_L = \sum_{h=0}^{L} N'_h c_h,$$

where $N'_0 = 1$ and $N'_h = (n-1)n^{h-1}$. Hence the invariant matrices form a commutative algebra of dimension $L+1$, spanned by the orthogonal projections P_k onto U_k.

Proof. $\mathrm{Aut}(T_L)$ acts transitively on ordered pairs of leaves with a given LCA height. It is transitive on leaves, and the stabiliser of a leaf i is transitive on the leaves at LCA height h from i. This gives $C_{ij} = c_{h(i,j)}$. Now fix $u \in U_k$ and a leaf i. Leaves j with $h(i,j) \le k$ lie in i's height-k block, where $u \equiv u_i$; there are N'_h of them at each $h \le k$. Leaves with $h(i,j) = k+1$ lie in the rest of i's height-$(k{+}1)$ block, where u sums to $-n^k u_i$. For $h(i,j) > k+1$, c is constant on whole height-$(k{+}1)$ blocks, on which u sums to zero. So $(Cu)_i = \big(\sum_{h \le k} N'_h c_h - n^k c_{k+1}\big) u_i$. □

Corollary 2.4 (learning decouples by scale)**.** *For linear regression on inputs with $\mathrm{Aut}(T_L)$-invariant covariance C, gradient flow $\dot\beta = -(C\beta - b)$ satisfies $P_k(\beta_t - \beta^*) = e^{-\sigma_k t} P_k(\beta_0 - \beta^*)$. The $L{+}1$ scales are learned independently, at rates σ_k. The preconditioner $\Pi = \sum_k \sigma_k^{-1} P_k$ is exact Newton, costs $O(n^L)$ to apply (one Haar pass), and is specified by $L{+}1$ numbers. Those numbers can be estimated by pooling over the orbit: a single input supplies $n^L N'_h$ leaf pairs for each c_h.*

Example 2.5 (broadcast data; the parameter $q = n\lambda^2$)**.** Let the leaves be ± 1 spins from a binary symmetric broadcast on T_L with edge correlation λ. Then $c_h = \lambda^{2h}$ and, with $q = n\lambda^2$,

$$\sigma_k = 1 + \big(1 - \tfrac{1}{n}\big) \sum_{h=1}^{k} q^h - \lambda^2 q^k.$$

If $q < 1$, all σ_k are $\Theta(1)$ and the problem is well conditioned. If $q > 1$, then $\sigma_k \approx \lambda^2 \frac{n-q}{q-1} q^k$ grows geometrically. Coarse scales are learned exponentially faster than fine ones (a *spectral bias across scales*), and the condition number is $\Theta(q^L)$. The same q is the Kesten–Stigum parameter that decides whether the root can be reconstructed (§5).

Checked numerically: eigenvalues of random invariant matrices match the formula to 10^{-14} for $(n, L) \in \{(2,4), (3,3), (4,2)\}$.

3 Discrete grids: automata, minimality, and the "copy" idea

Let Σ and $\mathcal{X}$ be finite. An SRG is then a deterministic bottom-up tree automaton on complete n-ary trees with a height-independent transition f. This identification makes the "if histories are similar, copy the grid" idea exact.

Definition 3.1 (context equivalence)**.** Fix a target $\phi : \Sigma^{[n]^L} \to \mathcal{Y}$. For block contents $s, s' \in \Sigma^{[n]^h}$ put $s \equiv_h s'$ iff $\phi(\sigma[w \leftarrow s]) = \phi(\sigma[w \leftarrow s'])$ for every node w of height h and every input σ (here $\sigma[w \leftarrow s]$ overwrites w's block). Let $k_h(\phi)$ be the number of classes. Two grids are equivalent iff no context, anywhere, can tell them apart.

Theorem 3.2 (Myhill–Nerode for SRGs)**.**

1. *$\equiv_h$ is a congruence: if $s_i \equiv_h s'_i$ for all i, then $(s_1, \ldots, s_n) \equiv_{h+1} (s'_1, \ldots, s'_n)$.*
2. *If a (possibly scale-untied) SRG computes $\phi = r \circ F_L$, then $F_h(s) = F_h(s')$ implies $s \equiv_h s'$. Hence it has at least $k_h(\phi)$ reachable states at height h.*
3. *The quotient SRG with states $\Sigma^{[n]^h} / \equiv_h$ and $f_h([s_1], \ldots, [s_n]) = [(s_1, \ldots, s_n)]$ computes ϕ, attains every bound in (2), and every SRG for ϕ with reachable states maps homomorphically onto it. It is the unique minimal SRG.*
4. *With scale tying: $\max_h k_h \le \min |\mathcal{X}| \le \sum_{h=0}^{L} k_h$.*

Proof. (1) Replace children one at a time. Each replacement swaps a height-h block inside some context, which by hypothesis leaves ϕ unchanged. (2) Equal states at w give equal states at every ancestor, whatever the context. (3) f_h is well defined by (1). By induction the quotient's state at w is the class of $\sigma|_w$, and $\equiv_L$ is equality of ϕ. The homomorphism sends a state to the class of any preimage, which is well defined by (2). (4) The lower bound is (2). For the upper bound, use the disjoint union of the height-wise quotients. □

Example 3.3 (which tasks are "self-similar")**.** Global parity and recursive majority have $k_h = 2$ at every height, and tied f with two states suffices. *Global* majority (over all n^L leaves) has $k_h = n^h + 1$ for all $h < L$, which is exponential in height. Every pair of block counts is separated by some context because $n^h \le n^L/2$. The SRG's natural task class is therefore the tasks whose *context rank* k_h stays bounded across scales. *Checked by brute force:* $(2, 2, 2, 2)$, $(2, 2, 2)$ *and* $(2, 4, 2)$ *respectively, and $k_h \le 3$ for a random 3-state tied automaton.*

Copying grids is lossless exactly on $\equiv_h$-classes. Replacing grid v by grid u (reusing u's outside state) never changes any output if and only if their contents are context-equivalent. This is the formal content of "replace one grid with another": the minimal SRG is the quotient of all grids by indistinguishability in every context.

Proposition 3.4 (memoized cost)**.** *Let f be scale-tied with finite $\mathcal{X}$. (i) Over* all inputs of all depths*, at most $|\mathcal{X}|^n$ distinct evaluations of f are ever needed, because f is a table. (ii) On an input given as a hash-consed DAG with $S(\sigma)$ distinct subtrees, memoized evaluation costs $O(S(\sigma))$ lookups. $S(\sigma)$ can be $O(L)$ for highly regular inputs, against $N_L \sim n^{L-1}$ nodes. (iii) A scale-untied SRG needs separate tables per height. Only tying allows a configuration seen at height h to be reused at height $h' \ne h$: if $C_u = C_v$ then $x_u = x_v$ whatever the heights and positions (§7).*

Remark 3.5 (Hashlife is this idea)**.** Gosper's Hashlife [11] simulates a radius-1 cellular automaton on a quadtree. By the light cone, the map Ψ_k from a 2^k-block to its central 2^{k-1}-block after 2^{k-2} steps is well defined. Ψ_k is built from 13 applications of Ψ_{k-1} to sub-blocks by the *same rule at every level*, and nodes are memoized by content. This is a (slightly generalized) scale-tied SRG in which the outside state is "the future of my centre", and it achieves exponential speedups on regular patterns. It is a working existence proof that the "copy identical grids" idea, combined with scale tying, yields exponential savings.

Learning. Learning a discrete SRG from data means learning a tree automaton. The idea of merging states whose contexts agree is the basis of provably correct non-gradient algorithms: identification in the limit by state merging [9], and polynomial-time learning of regular tree languages with membership and equivalence queries [19, 7]. The notes' "history coupling" is, in this language, *state merging*.

3.1 Histories: the temporal version

In the temporal SRG, grid w has a history $H_w(t)$. Call two histories *causally equivalent* if they induce the same conditional law of the future configuration process. The quotient is the minimal sufficient statistic of the past for the future: the causal states, or ε-machine [5, 21]. It is the temporal twin of Theorem 3.2, with "every context" replaced by "every future".

Proposition 3.6 (what copying histories does and does not preserve)**.** *Suppose grids u, v share dynamics (identical substrate) and have causally equivalent histories. Then substituting u's predictive state for v's is* exact for every marginal prediction about v. *However, generating v's future* by copying u's trajectory *gives the right marginal law for v while introducing an artificial dependence between u and v. Joint statistics of (u, v) can then be wrong, unless the true futures are conditionally independent given the histories and the copy uses fresh randomness.*

Proof. The predictive law of v depends on its history only through its causal state. For the joint claim, take two independent fair coins with identical (empty) histories: copying makes them perfectly correlated. □

So "enforce coupling in future trajectories" is safe for predicting each grid and unsafe for predicting how grids co-vary. The exact deterministic case is Hashlife.

4 Continuous grids: the scale gain κ

Let $\mathcal{X}$ be a normed space, f_θ differentiable with $J_i = \partial f/\partial c_i$, and define the *scale gain*

$$\kappa = \sup_c \sum_{i=1}^{n} \|J_i(c)\|_{\mathrm{op}}, \qquad B = \sup_c \|\partial_\theta f_\theta(c)\|_{\mathrm{op}}.$$

By the mean value inequality, f is κ-Lipschitz in the max-over-children norm.

Proposition 4.1 (approximate copying budget)**.** *Suppose that at every internal node of height h the computed value is replaced by any value within ε_h of $f(\tilde{C}_w)$, where $\tilde{C}_w$ are the (possibly already perturbed) children, for example a cached state of a nearby configuration. Then $\|\tilde{x}_w - x_w\| \le \sum_{j=1}^{\mathrm{ht}(w)} \kappa^{\mathrm{ht}(w)-j}\varepsilon_j$.*

Proof. $e_h \le \varepsilon_h + \kappa e_{h-1}$ with $e_0 = 0$. □

If $\kappa < 1$, tolerances can grow like $\kappa^{-(L-h)}$ towards the leaves. Aggressive approximate copying is safest exactly where the grids are most numerous. If $\kappa > 1$ (chaotic scale dynamics), only exact copying is safe, which is the discrete regime of §3.

Proposition 4.2 (anatomy of backprop through scale)**.** *For a loss $\ell(x_\epsilon)$, $\nabla_\theta \ell = \sum_{h=1}^{L} G_h$ with $G_h = \sum_{w \in V^{(h)}} \partial_\theta f(C_w)^\top \delta_w$, where $\delta_\epsilon = \nabla \ell$ and $\delta_{wi} = J_i(C_w)^\top \delta_w$. Moreover*

$$\sum_{w \in V^{(h)}} \|\delta_w\| \le \kappa^{L-h} \|\nabla \ell\|, \qquad \|G_h\| \le B\, \kappa^{L-h} \|\nabla \ell\|.$$

Proof. $\sum_i \|\delta_{wi}\| \le \sum_i \|J_i\| \|\delta_w\| \le \kappa \|\delta_w\|$; iterate, then sum over the level. □

Corollary 4.3 (scale starvation)**.** *If $\kappa < 1$, the bottom m levels contain a fraction $\approx 1 - n^{-m}$ of all grids but contribute at most $B\|\nabla \ell\| \kappa^{L-m}/(1-\kappa)$ to the gradient of the shared parameter. Backprop trains f mainly on the few top-level configurations and discards the statistical wealth at the bottom. Local rules (§6) weight every grid equally.*

Remark 4.4 (criticality)**.** κ is a worst case. For typical, weakly correlated Jacobians, the mean-field analogue replaces κ by $\chi = \mathbb{E} \sum_i \|J_i^\top u\|^2 / \|u\|^2$ in $\sum_{w \in V^{(h)}} \|\delta_w\|^2 \approx \chi^{L-h} \|\nabla \ell\|^2$. Backprop is balanced across scales only near $\chi = 1$, the tree analogue of the edge of chaos [20]. Because f is tied, criticality must hold at the *fixed-point distribution of states* that f itself induces deep in the tree (its renormalization-group fixed point), not merely at initialization.

The same κ therefore governs forward robustness (copying) and backward credit assignment. The regime where copying is cheap ($\kappa < 1$) is exactly the regime where backprop starves the bottom levels.

5 Self-similar processes: where the symmetry pays off

Up to now, symmetry has been a property of the architecture. It pays off statistically when the *process* generating the percepts is itself self-similar.

Definition 5.1 (self-similar hierarchical process, SSHP)**.** Let $|\mathcal{X}| = k$ and fix a root prior π. Draw $z_\epsilon \sim \pi$, and for each node draw $(z_{w1}, \dots, z_{wn}) \sim K(\cdot \mid z_w)$ with the *same* kernel K at every node. The leaves $\sigma_w = z_w$ are observed. In the *broadcast* (CI) case, $K(a \mid z) = \prod_i M_i(a_i \mid z)$ with column-stochastic $M_i \in \mathbb{R}^{k \times k}$.

Theorem 5.2 (exact inference is an SRG)**.** *Let $\beta_w(z) = P(\sigma|_w \mid z_w = z)$. Then $\beta_w = \Phi(\beta_{w1}, \dots, \beta_{wn})$ with $\Phi(\beta_1, \dots, \beta_n)(z) = \sum_a K(a \mid z) \prod_i \beta_i(a_i)$, which is $\bigodot_i M_i^\top \beta_i$ in the CI case. Leaves have $\beta_w = \mathbf{e}_{\sigma_w}$. Because Φ is multilinear, the normalized messages obey the tied recursion $x_w = f(C_w)$ with $f = \text{normalize} \circ \Phi$ on the simplex. The Bayes posterior of the root is $r(x_\epsilon) \propto \pi \odot x_\epsilon$. The downward (context) messages $\gamma_{wi} = M_i\big(\gamma_w \odot \bigodot_{j \neq i} M_j^\top \beta_{wj}\big)$ are also height-independent.*

Proof. Conditional independence of subtrees given z_w (standard belief propagation). □

So for an SSHP, the tied SRG is not merely a plausible inductive bias: its cell map is the Bayes-optimal one, a tied sum–product cell.

Theorem 5.3 (identifiability from the bottom, and labels from self-similarity)**.** *Consider a CI SSHP with $n \ge 3$, invertible M_i, and height-1 and height-2 latent marginals with full support.*

1. *The law of a* single bottom grid, *$\sum_z p_1(z) \bigotimes_i M_i[:, z]$, determines $(p_1, M_1, \dots, M_n)$ up to one common permutation P of the latent labels (Kruskal [15, 1]). Jennrich's simultaneous diagonalization recovers them efficiently.*

2. *Let R range over permutation matrices. If there is no pair $(R \neq I, S)$ with $R^\top M_i S = M_i R$ for all i, then the laws of height-1 and height-2 grids determine P. The kernel is then identified* including its labels *with no labelled data. This condition fails only on a Lebesgue-null set of kernels.*

Proof. (1) The children are CI given the parent, and three invertible factors have Kruskal ranks summing to $3k \geq 2k+2$. (2) Level 1 gives $\hat{M}_j = M_j P$. Child i of a height-2 grid emits the leaf tuple of a height-1 grid through the channel AM_i, where $A = M_1 \odot \cdots \odot M_n$ (column-wise Khatri–Rao) has full column rank. Kruskal at height 2 gives AM_iQ for an unknown permutation Q. Since $\hat{A} = AP$ is known, we obtain $D_i = \hat{A}^+ AM_iQ = P^\top M_i Q$. A candidate labelling P' asserts the kernel M_jR, where $R = PP'^\top$. Self-similarity requires the height-2→1 channel in leaf labels, $P'D_iQ'^\top$, to equal M_iR for some Q'. That is $R^\top M_i S = M_i R$ with $S = QQ'^\top$. The truth $R = I$ satisfies this with $S = I$. For each fixed $(R \neq I, S)$ the condition is a linear subspace; it excludes some elementary matrix E_{ab}, so it is proper. The column-stochastic matrices span the matrix space, so the subspace meets them in a null set. There are finitely many (R, S). □

Remark 5.4. The symmetric channel $M = aI + b\mathbf{1}\mathbf{1}^\top$ is degenerate ($S = R^2$ works). Indeed, "child copies parent" and "child copies r(parent)" then produce the same leaf law. Top-down information (labels) is needed exactly to break this residual ambiguity, which consists of at most $k!$ hypotheses. *Checked: the demo kernel is non-degenerate, and likelihood selection among the $k!$ candidates recovers the true labelling.*

Theorem 5.5 (bottom statistics: effective sample size)**.** *Let λ be the largest maximal (Hirschfeld–Gebelein–Rényi) correlation of a parent–child pair, $q = n\lambda^2$, $N_1 = n^{L-1}$, and g any function of a bottom configuration with* $\operatorname{Var} g(C_w) \leq V$. *For $\hat{g} = N_1^{-1} \sum_{w \in V^{(1)}} g(C_w)$,*

$$\operatorname{Var} \hat{g} \leq \frac{V}{N_1}\Big[1 + \big(1 - \tfrac{1}{n}\big) \sum_{j=1}^{L-1} q^j\Big].$$

Hence $N_{\text{eff}} = V/\operatorname{Var} \hat{g}$ is at least $\frac{1-q}{1-q/n}\, n^{L-1}$ if $q < 1$, at least n^{L-1}/L if $q = 1$, and at least $c_q\, \lambda^{-2(L-1)}$ with $c_q = \frac{q-1}{(1-1/n)q}$ (asymptotically in L) if $q > 1$. In every regime it grows exponentially with depth, whereas root supervision provides one label per example.

Proof. For $u \neq v$ at height 1 with LCA at height $j+1$, $g(C_u) - z_u - \cdots - z_{\text{lca}} - \cdots - z_v - g(C_v)$ is a Markov chain with $2j$ latent edges. Maximal correlation is submultiplicative along Markov chains [23], so $|\operatorname{Cov}| \leq \lambda^{2j} V$. Each u has $(n-1)n^{j-1}$ such partners. Sum. □

Checked: for the binary broadcast the bound is within a factor 1.2–1.6 *of the exact variance for $q \in \{0.48, 1, 1.92\}$ and L up to* 12*, and the exact formula matches Monte Carlo.*

Theorem 5.5 is pessimistic for learning the *kernel*. The next result is the one that matters.

Theorem 5.6 (one example contains n^{L-1} independent kernel samples)**.** *Conditioned on the height-1 latents $Z_1 = (z_w)_{w \in V^{(1)}}$, the bottom configurations are* independent*, with $C_w \sim K(\cdot \mid z_w)$. Consequently the empirical moment tensor is*

$$\hat{T} = \sum_z \hat{p}_z \bigotimes_i M_i[:, z] + E, \qquad \mathbb{E}\big[\|E\|_F^2 \,\big|\, Z_1\big] \leq \frac{1}{N_1},$$

where $\hat{p}$ is the empirical histogram of Z_1. All correlation between bottom grids enters through $\hat{p}$, and it leaves the rank-k factors unchanged*. Any factor-recovery method that allows arbitrary positive weights (Jennrich, tensor power iteration, EM with free weights) therefore sees noise of order $N_1^{-1/2} = n^{-(L-1)/2}$, independently of q. The only requirement is that $\min_z \hat{p}_z$ stays bounded away from zero.*

Proof. Tree Markov property. $E = N_1^{-1} \sum_w (\mathbf{e}_{C_w} - \mu_{z_w})$ is a sum of conditionally independent, conditionally centred terms of norm at most 1. □

Demonstration. We use $k = n = 3$ and a generic diagonally dominant kernel with $\lambda_2 = 0.72$ ($q = 1.56$, above Kesten–Stigum). The task is to classify the root latent. The Bayes oracle scores 0.724, census (leaf majority) 0.527, and chance 0.333.

method	data	root accuracy
bottom-level learning (Jennrich + EM on height-1 grids), labels pinned by Thm. 5.3; zero labels		
one tree of depth 6 / 7 / 8 / 9	0 labels	0.47 / 0.40 / 0.706 / **0.722**
labels correctly pinned (of 5 runs)		3 / 2 / 5 / 5
kernel error (up to relabelling), same trees		0.076 / 0.044 / 0.025 / 0.013
20 / 50 / 100 trees of depth 5	0 labels	0.634 / 0.706 / 0.718
backprop from root labels, exact tied BP architecture (best of 3 inits)		
10 / 30 / 100 / 300 / 1000 trees	labelled	0.515 / 0.655 / 0.712 / 0.723 / 0.662
backprop from root labels, generic neural tied SRG		
100 / 1000 / 10000 trees	labelled	0.553 / 0.618 / 0.684

Kernel error from a single tree falls by about $1/\sqrt{3}$ per level of depth, the $N_1^{-1/2}$ rate of Theorem 5.6. An ablation comparing $81U$ bottom grids taken from U trees with $81U$ *independent* grids gives the same kernel error (e.g. 0.146 vs. 0.171 at $U = 1$; 0.017 vs. 0.013 at $U = 100$), as the theorem predicts. Pinning the labels (Thm. 5.3(2)) uses height-2 statistics, which are fewer and correlated, and so needs more depth. The kernel learned at the bottom level alone, then deployed at every level, reaches the oracle without a single label or backward pass. Backprop through the *same* ideal architecture needs on the order of 100–300 labels and suffers from local optima (the non-monotone 1000 entry). A generic network is far worse. These are toy-scale numbers. They illustrate the theorems and are not benchmarks.

Remark 5.7 (Kesten–Stigum duality)**.** For the binary broadcast, the root is reconstructible from the leaves as $L \to \infty$ iff $q > 1$ [13, 8, 18]. The same q appears in Theorem 5.5 (redundancy of bottom statistics) and in Example 2.5 (conditioning across scales). Below the threshold, bottom statistics are almost independent but the top-level task disappears. Above it, the task is solvable and plain averages become correlated. The kernel itself (Thm. 5.6) is protected from this trade-off.

6 Learning rules that exploit the symmetry

In a tied SRG, one input presents the *same* local problem "map a configuration to an outside state" N_L times. Backprop sums the corresponding gradients through products of up to L Jacobians (Prop. 4.2). Each rule below exploits the repetition differently.

6.1 Frozen-input learning and scale equilibria

Give every grid a local loss $\ell_w(f_\theta(C_w))$, for example a self-supervised prediction by the observer. The *local rule* updates θ with $g_{\mathrm{loc}} = \sum_w \partial_\theta \ell_w(f_\theta(C_w))$, treating the inputs C_w as constants (stop-gradient). There is no backward pass across scales, all N_L grids update in parallel, and memory is $O(1)$ per grid.

Definition 6.1 (scale equilibrium)**.** θ^* is a *scale equilibrium* if $\theta^* \in \arg\min_\theta \sum_w \mathbb{E}\, \ell_w(f_\theta(C_w(\theta^*)))$: the shared strategy is a best response to the inputs it itself produces. This is a symmetric equilibrium of the game whose players are the grids and whose common strategy is the notes' S.

Fixed points of the local rule are scale equilibria. They generally differ from minimizers of the end-to-end loss, just as TD fixed points differ from minimizers of the value error. They coincide under *local realizability*: if $\ell_w \geq 0$ and $\ell_w = 0$ almost surely at θ^* for every w, then θ^* is both a global minimizer and a scale equilibrium. The SSHP results of §5 are a case where local, moment-based estimation is provably consistent.

6.2 Scale-Bellman bootstrapping

Suppose the observer's target is consistent across scales: $y^*_w = g(y^*_{w1}, \ldots, y^*_{wn})$ for a known aggregator g (counts, maxima, logical rules, averages). Ground truth is available only at small scales, which is cheap. Train with the one-level loss $\|r(x_w) - g(\overline{r(x_{w1})}, \ldots, \overline{r(x_{wn})})\|^2$, where the bar denotes a stopped gradient, exactly as a TD target.

Proposition 6.2. *If the consistency residual is at most ε at every internal node, the leaf error is at most ε_0, and g is Λ_g-Lipschitz in the max norm, then $|r(x_w) - y^*_w| \leq \Lambda_g^h \varepsilon_0 + \varepsilon \sum_{j<h} \Lambda_g^j$ at height h. For averaging aggregators ($\Lambda_g = 1$) the error grows only linearly in depth, as $\varepsilon_0 + h\varepsilon$. This is the scale analogue of the $1/(1-\gamma)$ horizon factor.*

Proof. $e_h \leq \varepsilon + \Lambda_g e_{h-1}$. □

6.3 Node-parallel evolution strategies

Untie temporarily: give node w the copy $\theta_w = \theta + s\xi_w$ with $\xi_w \sim N(0, I)$ i.i.d., run a forward pass with $+\xi$ and one with $-\xi$, and set $\hat{g} = \frac{1}{2sN_L} \sum_w (\ell^+_w - \ell^-_w)\xi_w$.

Proposition 6.3. $\mathbb{E}\hat{g} = \frac{1}{N_L} \sum_w \partial_{\theta_w} \ell_w \big|_{\theta_w \equiv \theta} + O(s^2)$. *This is exactly the frozen-input gradient g_{loc}/N_L, computed with two forward passes, no backward pass, and a population of size N_L at no extra cost.*

Proof. ℓ_w depends on its own copy and on its descendants' copies. Expand to first order: since $\mathbb{E}\,\xi_u \xi_w^\top = \delta_{uw} I$, only the own-copy term survives in the mean. Descendants add variance but no first-order bias, and antithetic pairs cancel the even-order terms. □

Descendant cross-terms add variance scaled by path sensitivities, which again are bounded by powers of κ. Heuristically, the relative error behaves like $\sqrt{\dim \theta / N_L}$, which makes ES practical once $N_L \gg \dim \theta$. The same machinery supports *untie–diversify–select–retie*: keep different θ_w in different grids, score them with local losses, and copy the winners everywhere. Self-similarity guarantees that a copy which works in one grid is admissible in every grid. *Checked: with $N_L = 63$, the ES estimate is within 8% of the frozen-input gradient (Monte Carlo noise) and 65% away from the full backprop gradient, which itself differs from the frozen-input gradient by 63%.*

6.4 Summary of rules

rule	what it uses	guarantee
bottom-level moments + BP	N_1 CI copies of the kernel per example	identifiable, labels pinned by self-similarity (Thms. 5.3, 5.6)
frozen-input (local)	all N_L grids weighted equally	scale equilibrium; exact under local realizability
scale-Bellman	consistency of targets across scale	error $\leq \varepsilon_0 + h\varepsilon$ (Prop. 6.2)
node-parallel ES	N_L-sized population for free	unbiased local gradient to $O(s^2)$ (Prop. 6.3)
copy / merge	context or history equivalence	lossless iff equivalent; minimal machine (Thm. 3.2)
approximate copy	contraction $\kappa < 1$	root error $\leq \sum_h \kappa^{L-h}\varepsilon_h$ (Prop. 4.1)
scale preconditioning	L+1-dim commutant	exact Newton for invariant quadratics (Cor. 2.4)

7 Transfer, including "diagonal" transfer

Proposition 7.1. *In a position-shared, scale-tied SRG:*

1. *At the level of parameters, transfer between any two grids is total and blind to position and scale. Their quotient is a single vertex with n loops (Prop. 1.3), so "diagonal" transfer between non-ancestral grids at different heights is indistinguishable from vertical transfer.*
2. *Upward transfer is exact: by Lemma 1.2, $F_{h'}$ contains $n^{h'-h}$ copies of F_h.*
3. *At the level of* states*, tying is what makes diagonal transfer possible: if $C_u = C_v$, then $x_u = x_v$ for* any *heights and positions. A scale-untied SRG has no such property. Memo tables, merged states and coupled histories can therefore be shared across scales (Prop. 3.4(iii)).*

The genuinely new content of "diagonal transfer" in the notes is therefore state- or history-level transfer (§3), not parameter-level transfer. Self-similarity of the *input* decides how much of it there is. On a fractal input, every level can reuse the same few table entries.

8 Relation to prior work, and what is new

Most ingredients have precedents. Recursive and tree-structured networks and backprop through structure [10, 22] already use this kind of architecture. Weight tying across depth appears in Universal Transformers and deep equilibrium models [6, 3]. Local tied update rules appear in neural cellular automata [17]. Same-network-at-every-level part–whole hierarchies appear in GLOM [12]. The mapping between deep learning and the real-space renormalization group is due to [16], and a mutual-information RG learning rule to [14]. The Random Hierarchy Model [4] studies how backprop-trained CNNs learn hierarchical grammars, with rules drawn independently per level, i.e. *untied*. On the statistical side, reconstruction on trees [13, 8, 18] and spectral/tensor learning of latent trees [15, 1, 2] are also established. Hashlife [11] is the computational precedent for "copy identical grids". The idea that one algorithm is reused across the cortical hierarchy is Mountcastle's.

What these notes add is a single formal object that ties these threads together, plus the following specific results, which we have not seen stated in this form: the exact invariance group and completeness of exchangeable SRGs (Prop. 2.2); the $(L+1)$-dimensional scale spectrum and its link to $q = n\lambda^2$ (Thm. 2.3, Ex. 2.5); Myhill–Nerode minimality for scale-tied grids as the formal version of grid copying, and the observation that tying is what makes cross-scale copying valid (Thm. 3.2, Prop. 7.1); the single constant κ governing both copy error and backprop's scale starvation (Props. 4.1, 4.2); label identification by self-similarity (Thm. 5.3(2)); the conditional-independence argument that one deep example supplies n^{L-1} independent kernel samples (Thm. 5.6); and node-parallel ES as a backward-free estimator of the frozen-input gradient (Prop. 6.3). A literature search before publication is still advisable. Some of these may exist in scattered form.

9 Open problems

Open problem 9.1 (the gap between scale equilibria and backprop optima)**.** Bound the distance between the fixed point of frozen-input learning and the end-to-end optimum in terms of κ (or χ) and a measure of local realizability, in analogy with the TD fixed-point bound $\|V_{TD} - V^*\| \leq (1-\gamma^2)^{-1/2}\|\Pi V^* - V^*\|$.

Open problem 9.2 (tied Random Hierarchy Model)**.** In the RHM with rules *tied* across levels, show that bottom-level learning (clustering synonymous tuples by sibling co-occurrence) has sample complexity independent of L. Compare this with the $\propto m^L$ scaling reported for backprop-trained CNNs on the untied model [4]. A separation would be a clean "backprop is not optimal here" theorem. *Largely addressed in the companion note* ***srhm.pdf****.* There, the number of unlabelled examples needed falls like $s^{-(L-2)}$, down to one example, with zero labels. In the untied model every learner needs $\Omega(vm)$ labels. A lower bound for backprop itself remains open.

Open problem 9.3 (continuous Myhill–Nerode)**.** Define ε-context equivalence for real-valued SRGs and relate its covering numbers to the minimal dimension of $\mathcal{X}$ and to the cost of approximate memoization.

Open problem 9.4 (time $\times$ scale)**.** When do the causal states of a parent grid factor through the causal states of its children? (Sufficient: independent children processes.) Build a learned, stochastic Hashlife.

Open problem 9.5 (criticality without backprop)**.** The sibling ensemble at each height gives free estimates of Jacobian statistics. Design a local rule that keeps $\chi \approx 1$ at the self-induced fixed-point distribution.

Open problem 9.6 (lattice symmetry)**.** Extend the harmonic analysis of §2 from $S_n \wr \cdots \wr S_n$ to the position-aware group $\mathbb{Z}_b^d \wr \cdots \wr \mathbb{Z}_b^d$ of the lattice variant.

A Reproducibility

All checks run on a laptop CPU. `verify/checks.py` covers Props. 2.2 and 6.3, Thms. 2.3, 3.2 and 5.5, and runs in about 25 s. `verify/demo_sshp.py` runs the SSHP demonstration and backprop baselines, and `verify/demo_one_tree.py` the single-tree experiment (Thm. 5.6). Outputs are saved next to the scripts.

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